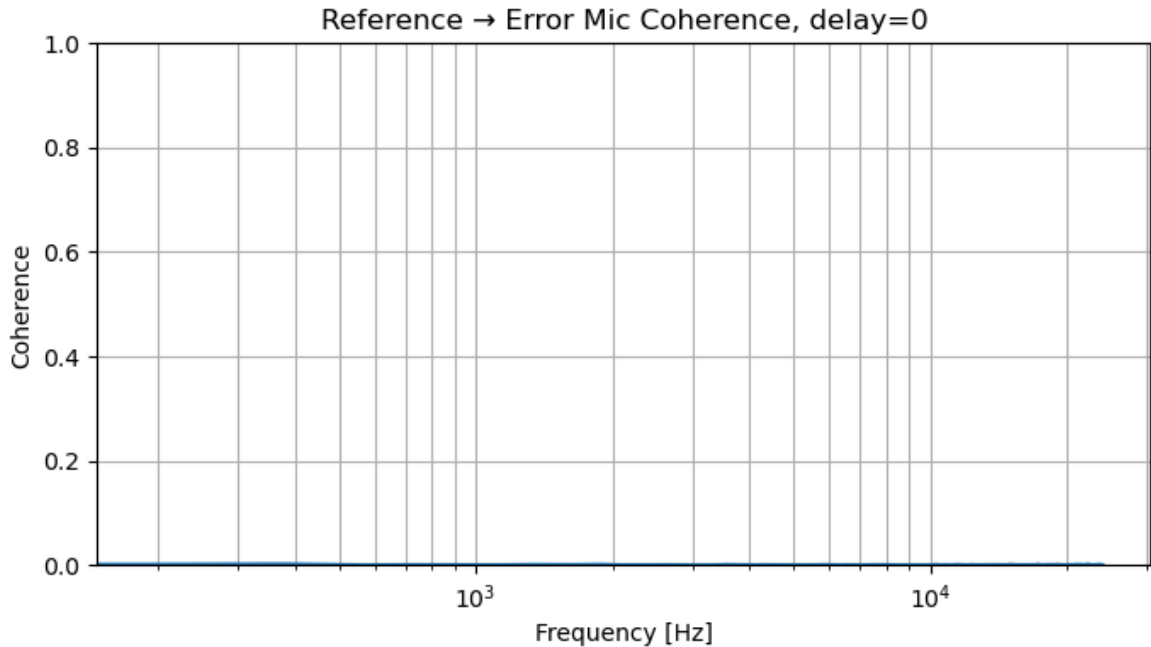
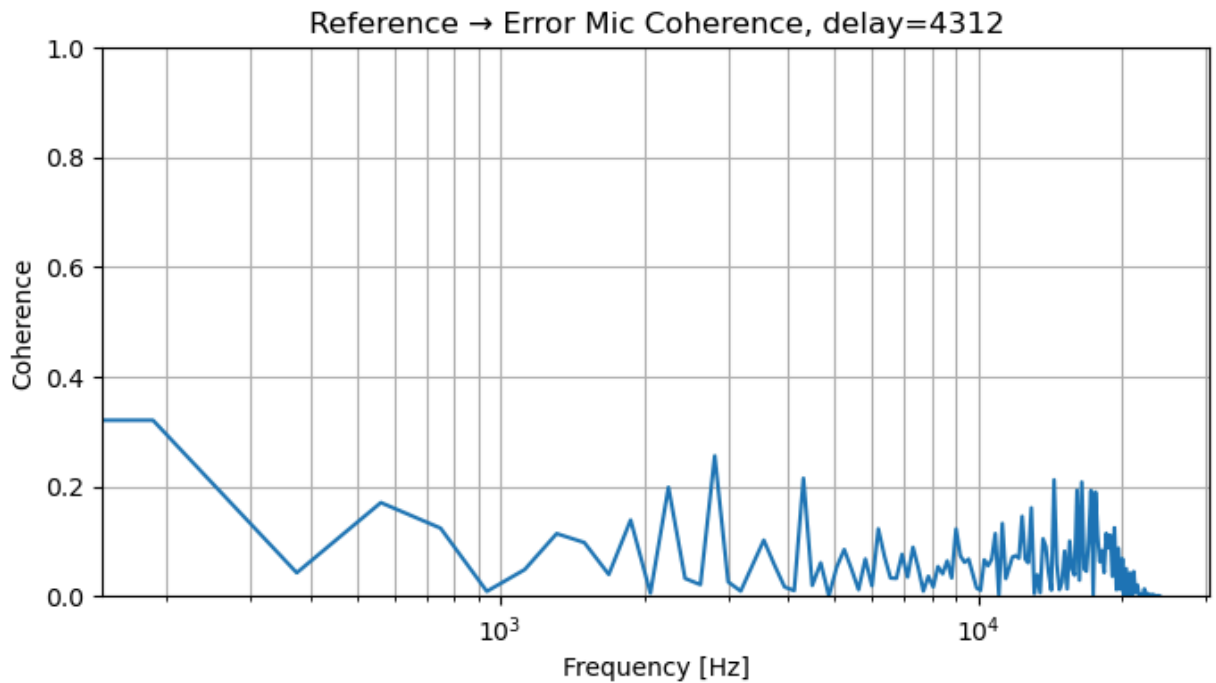


System delay:

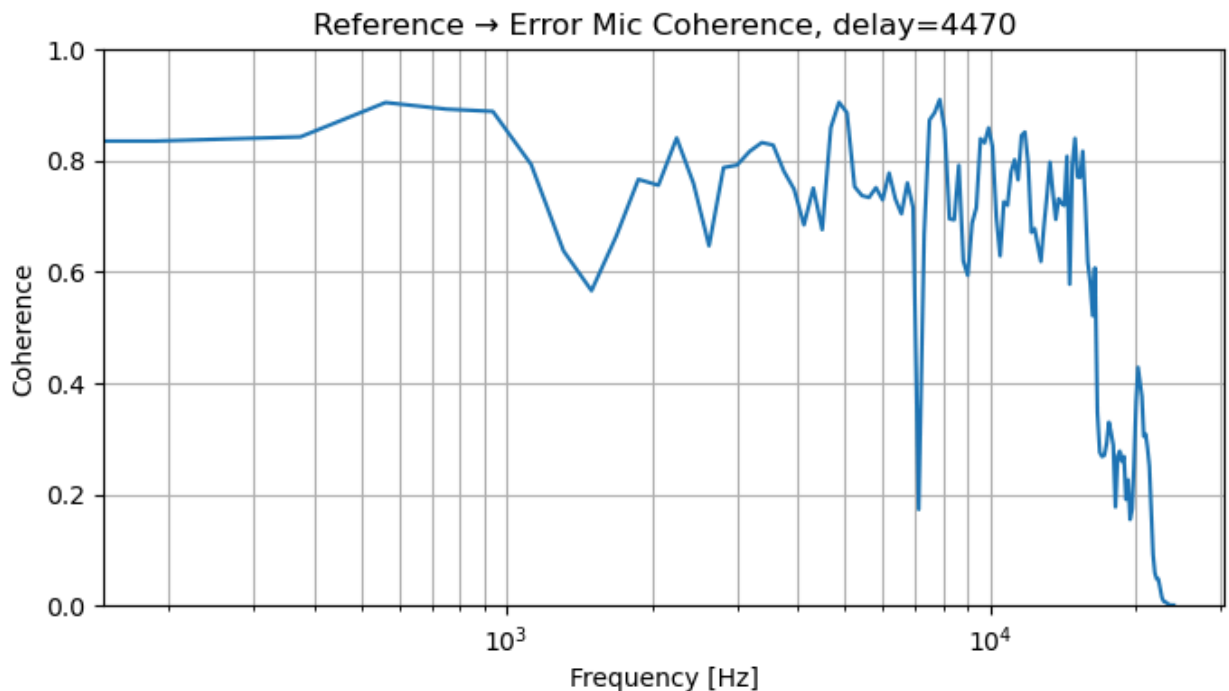
- The raw non-cancelled error mic recording has no coherence with the raw source



- The streaming function has a delay of **4312** samples!!! So a click sent from the output will be recorded by the mic 4312 samples later, or about 90ms
- When the error recording is shifted by that amount, there is more coherence



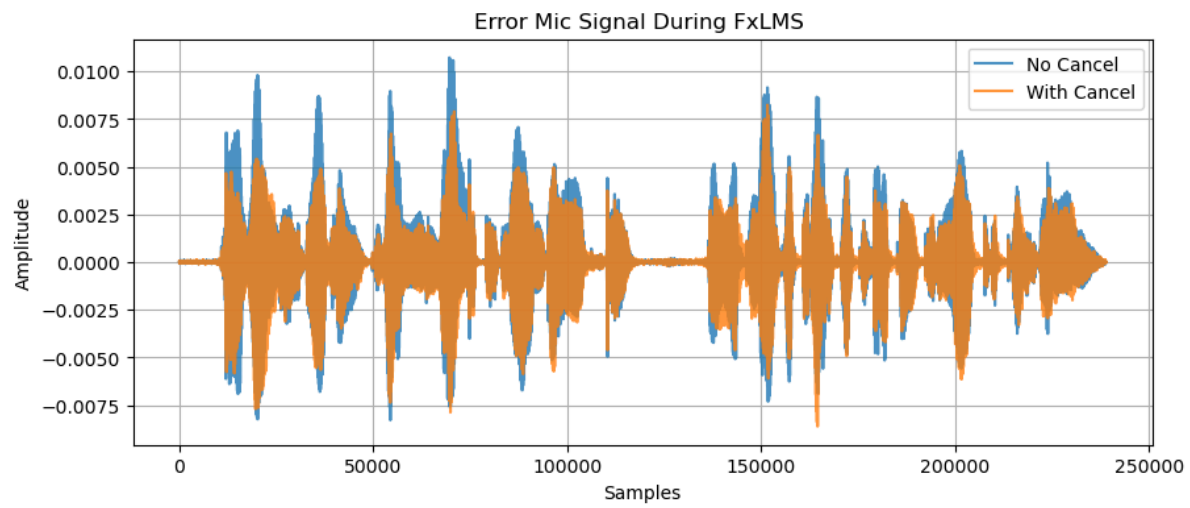
- And if we do a cross-correlation, the delay with most coherence it finds is ~4475 samples



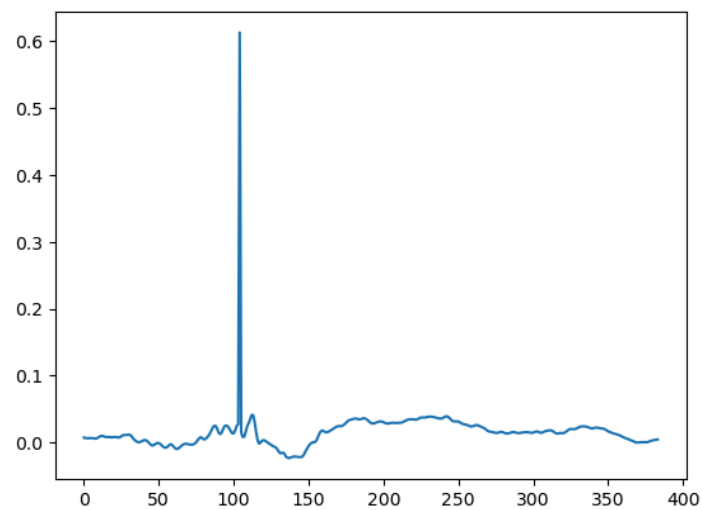
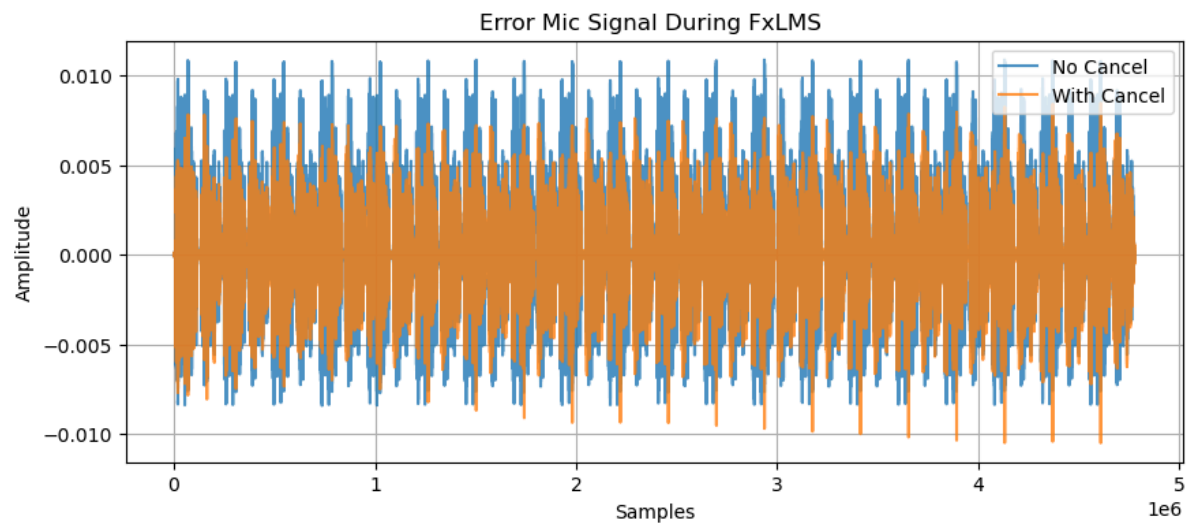
- Made the weight update use a delayed reference
- Even adding the delay to the callback didn't fully work, maybe some bugs?

System delay:

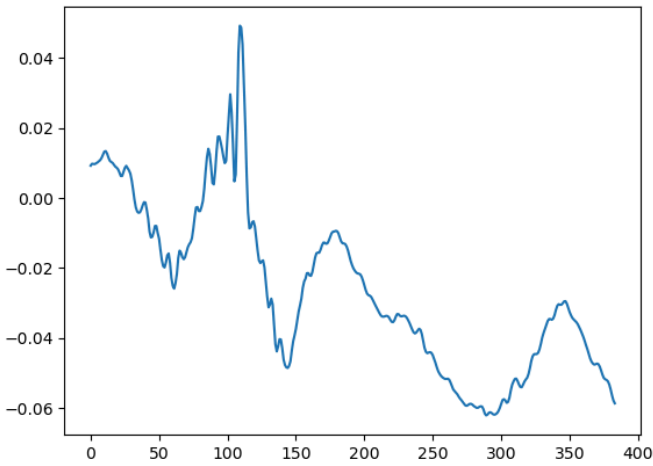
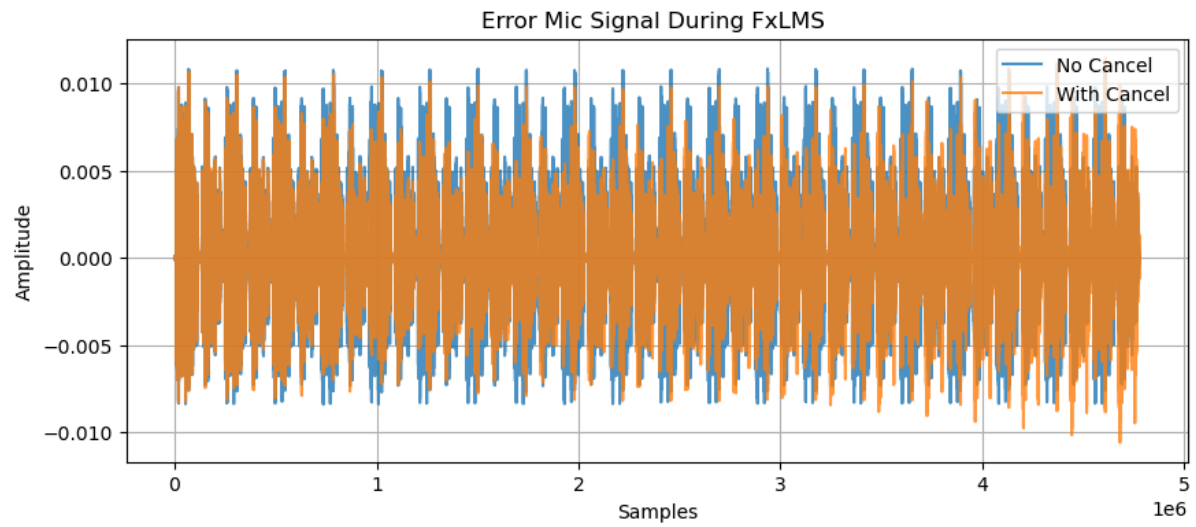
- I ran a fine grid search over seeding the filter with a delta function at different indices with +/-1 as the value. I learned with the 4312 delay instead of the cross-correlation
- Found the following 4 indices that actually got cancellation
 - d=96, sign=+1, c_gain=0.2 -> -1.25 dB
 - d=104, sign=+1, c_gain=0.2 -> -1.71 dB
 - d=136, sign=-1, c_gain=0.2 -> -1.01 dB
 - d=144, sign=-1, c_gain=0.2 -> -1.45 dB



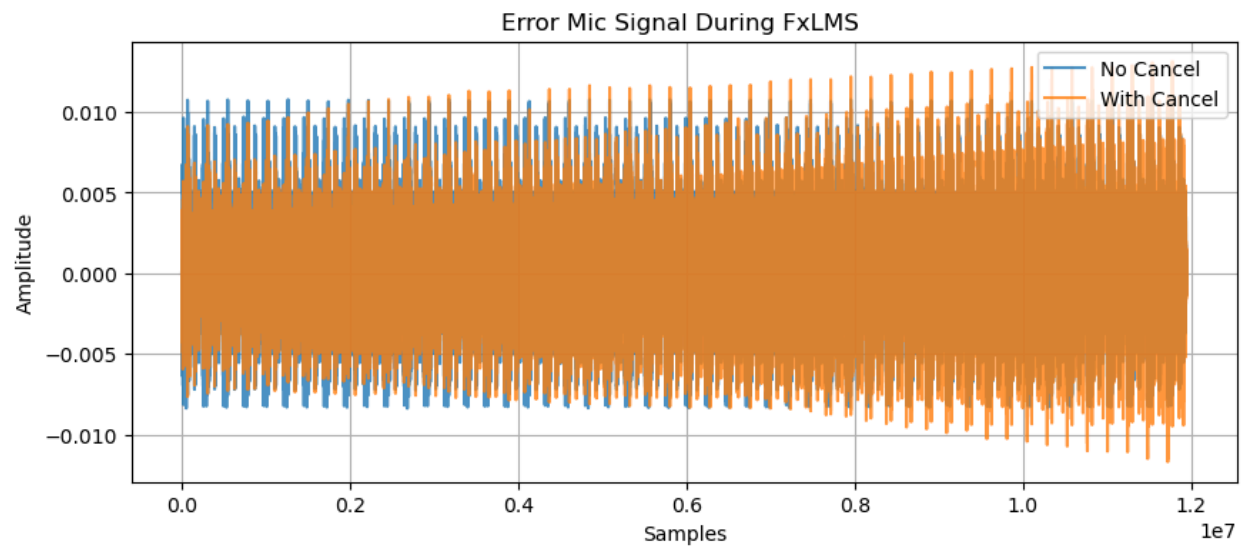
- Running fxnls on a seeded w at 104 improved! -2.7 dB reduction



- It also started learning when I didn't use the seed, -1.4 dB reduction

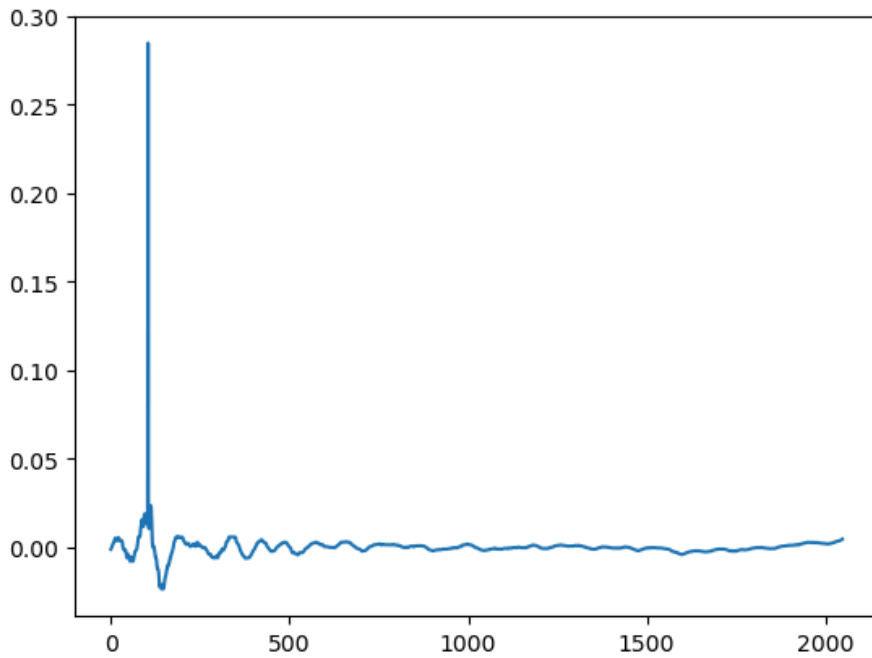
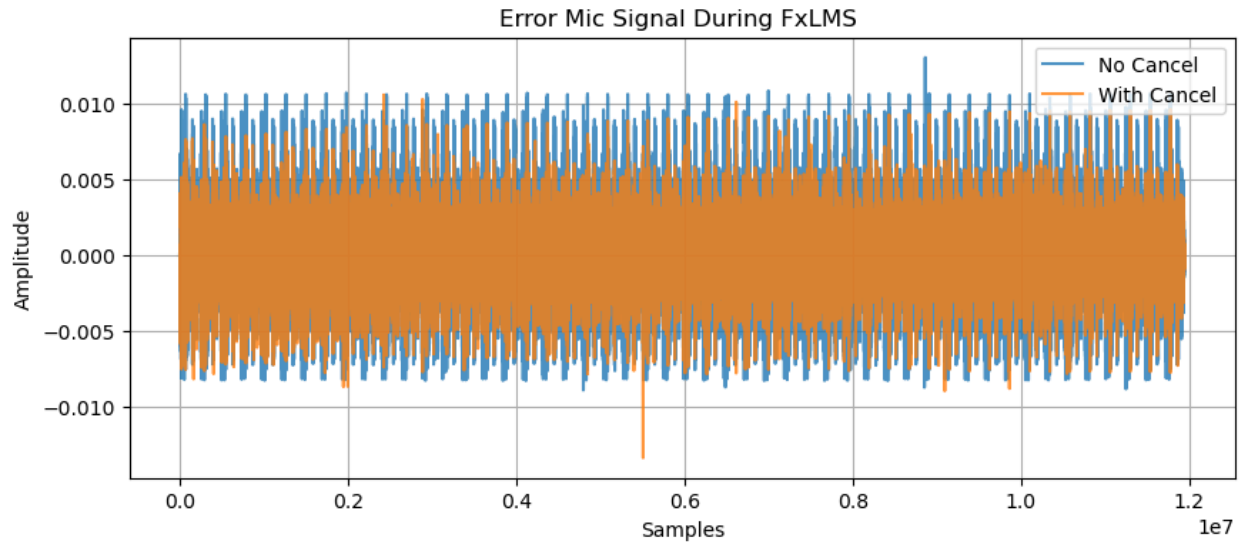


- For whatever reason, it didn't work when I used the cross-correlation lag

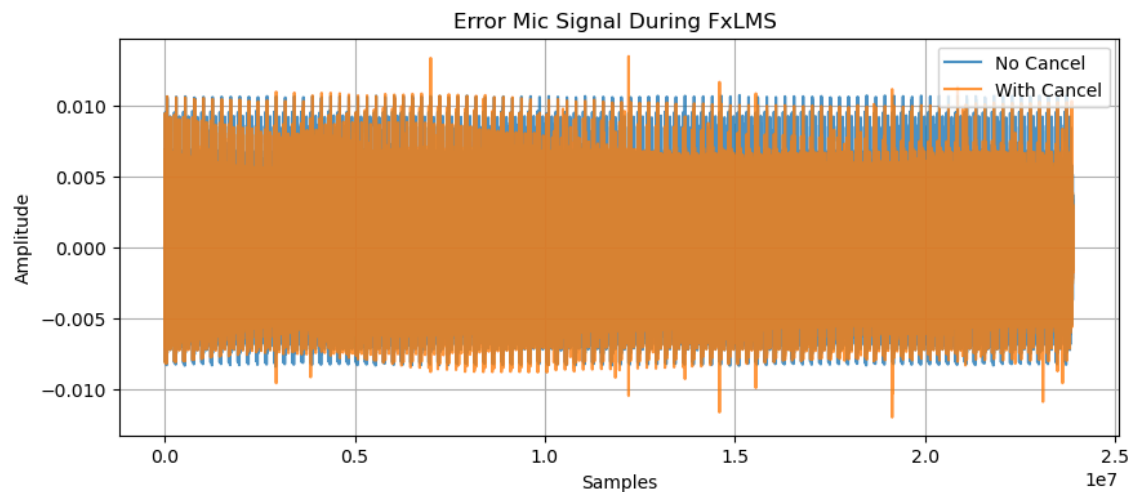


More iterations:

- The learning was not stable for more iterations
- If I used a longer filter, it was more stable. 2048 filter length got -3.6 dB reduction on 50 iterations when seeded

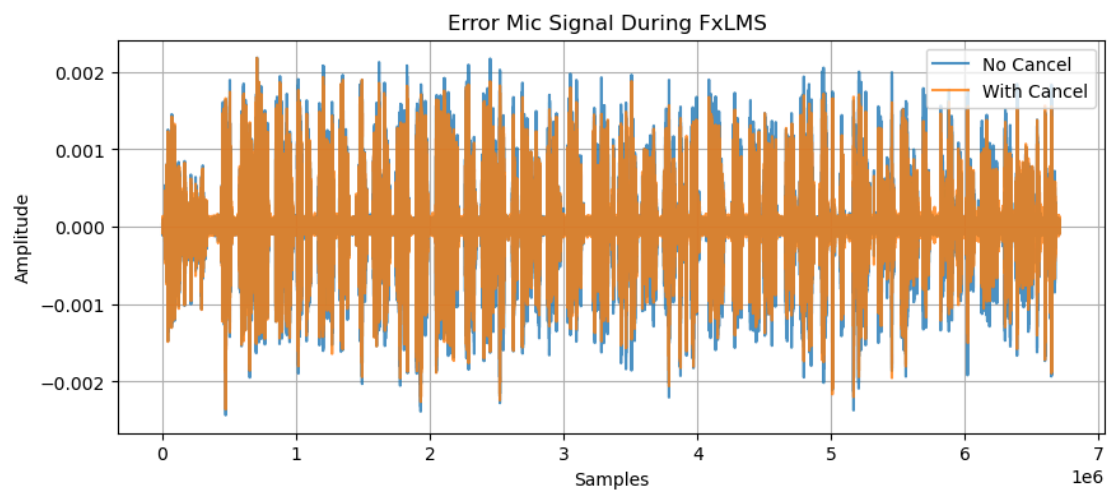


- Grid search over filter length and block size showed that block size doesn't matter much, but 2048 is the best filter length for 50 iterations
- I also tried 100 iterations with no seed, 2048 filter length. It got -2 dB reduction. Maybe a longer filter would be better?

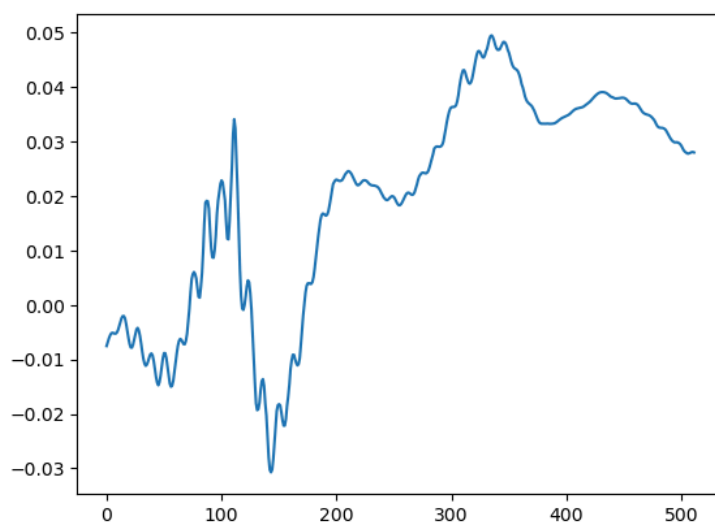


Other audio:

- I also tried it on the full arthur recording. It worked ok, but not as well. No seed here



- The filter shapes it learns are pretty weird at the end



- I tried learning a filter on the short arthur clip, then using that on the full arthur clip with no learning, and that worked pretty well, -1.6 dB reduction. So the algorithm does learn an applicable filter

